

Nameer Iqbal Ansari

RTL Engineer | ASIC | RISC-V

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PROFESSIONAL SUMMARY

RTL Design Engineer (FPGA/ASIC) at XCLR Technologies, specializing in RISC-V-based SoC development. Certified RISC-V Foundational Associate (RVFA) and Google Summer of Code (GSoC) 2022 contributor and 2023 mentor, with hands-on experience in SystemVerilog/Verilog based ASIC and FPGA designs, Lint/CDC/RDC sign-off, timing closure, and verification.

TECHNICAL SKILLS

- **RTL Design** — RISC-V | OOO Core | VLSU | JTAG | SerDes | PCIe | eFuse | UART | CDC | XBAR | Cache | AXI | APB | CHI | TileLink
- **EDA Tools** — VCS | Verdi | Xilinx Vivado | GTKWave | CoreAssembler | CoreConsultant | QuestaSim | Verilator | FusionCompiler | PrimeTime
- **Formal Verification** — JasperGold | SVA (SystemVerilog Assertions)
- **Static Analysis** — Synopsys SpyGlass (Lint, CDC, RDC)
- **Languages** — Verilog | SystemVerilog | SystemVerilog-OOP | UVM (basic) | C | C++ | RISC-V Assembly
- **Scripting** — Tcl | Shell | Makefile | CI/CD | Python | YAML (FuseSoC)
- **Version Control** — GitHub | GitLab
- **Operating System** — Linux | Windows

WORK EXPERIENCE

02/2025 – Present

RTL Design Engineer (FPGA/ASIC), XCLR Technologies (Pvt.) Ltd.

- **High-Speed I/O Subsystem (HSIOSS)** — Led design/verification (3 teams, 6 engineers) of a 32 Gbps SerDes + PCIe Gen5 subsystem at 1.5 GHz/100 MHz clocks, 2048-bit/64-bit AXI, 8-lane 32-bit each MAC and PHY PIPE. Built the verification plan firmware + SystemVerilog/UVM testbenches. Defined the verification plan chip-level firmware tests plus block-level SystemVerilog testbenches and modified/utilized existing UVM tbs within the IPs
- **RISC-V Vector Load/Store Unit (VLSU)** — Designed a 2048-bit RVV load/store unit interfacing to a 64-bit memory port, with AXI-ID tracking, address generation and translation, and data packing/unpacking logic supporting configurable element widths (8–64 bits) within a single vector register. Added multi-stage pipelining to reduce latency and cell count
- **IP Replacement Across 5 Subsystems** — Re-architected 5 subsystems by replacing Synopsys-sourced data/protocol converters, crossbars, CDC synchronizers, and AXI-to-native-memory bridges with PULP Platform IP. Utilized Synopsys CoreAssembler to generate a GUI-based integrated RTL, and also used CoreConsultant to generate standalone Synopsys IPs
- **Unified DFT/Functional JTAG Architecture** — Architected a single chip-level JTAG interface for two independent TAP controllers (DFT scan-based and functional RISC-V PULP debug) by muxing in between both TAP interfaces onto one physical port, enabling both production test and hardware debug through shared JTAG standard pins
- **Control-Register-to-AXI Bridge** — Designed and verified a protocol bridge between a 16-bit native SerDes control-register interface and a 64-bit AXI configuration bus, resolving CDC and 64-to-16-bit data-width conversion. Validated the design with block-level SV TB and system-level firmware testing after integration of the converter into the chip
- **512-bit eFuse Controller** — Implemented 32-bit AXI access by bridging the native eFuse memory array interface to an AXI-based register-access architecture, including command-driven read/write transaction handling. Performed SystemVerilog-based block-level testing and validated end-to-end chip-level functionality through QSPIS interface access testing
- **Chip-level low-power implementation** — PPA analysis enabled 80% of the chip in low-power mode via Synopsys IP + custom logic. Built a glitch-free clock-gating power-switch mechanism, safely handling in-flight AXI transactions via SLVERR/DECERR during power-state transitions

09/2023 – 02/2025

Associate Engineer, Xcelerium

- **12nm timing closure at 1.5 GHz of RISC-V-based core subsystem** — Achieved -0.01 worst negative slack (WNS) and reduced setup and hold (reg2out/in2reg/in2out) violations across 500+ violating paths by optimizing combinational logic, skew balancing, registering, floorplanning, and SDC constraint fixes
- **JTAG daisy Chain at Aries** — Designed and validated a daisy-chained JTAG debug architecture linking two PULP riscv-dbg TAP controllers across two subsystems, allowing either core to be halted and debugged independently while the other ran normally. Verified via SimJTAG (C DPI) in simulation and on AWS EC2 cloud FPGA
- **Lint, CDC, and RDC sign-off** — Accomplished 0 error/warning sign-off across 4 subsystems, each containing a minimum of 100 violations that are detected using Synopsys SpyGlass, ensuring digital designs are ready for physical design (PD)

- **RTL design & Digital IP integration** — Integrated digital components in the RTL designs including logic analyzers, **CDC** synchronizers, **AXI** to **APB** protocol converters, data width converters, crossbar interconnects, register files and **FSMs**
- **Technical Documentation & Design Reviews** — Authored comprehensive design documents, verification test plans, timing diagrams, and block diagrams to aid design reviews and knowledge transfer

03/2020 – 09/2023

Research Assistant, *Micro Electronics Research Lab (MERL)*

- **RV32I Core Design** — Built an open-source **RV32I CPU core** [↗](#) starting from scratch using SystemVerilog, simulated with **Verilator** using **FuseSoC**, and emulated on **Arty A7 FPGA** using **Xilinx Vivado**. Later integrated basic **UVM** and **JasperGold** support
- **Out-of-Order Upgrade to RV32I Core** — Added out-of-order execution: **2-wide fetch**, **RAT**-based register renaming, **ROB** for in-order commit, and reservation stations feeding **ALU/branch units** migrating the **pipeline** to a **dispatch/issue/commit** structure, **BOOM-inspired**
- **NOVA1 SoC Development** — Contributed to **NOVA1 System-on-Chip (SoC)** development and its **AWS EC2-based** emulation, and **Zyfer OS** bring-up
- **Multicore Cache Coherency & CHI Research** — Led the research of the **multicore architecture** and its **cache coherency** through the **OpenPiton** framework. Also researched converting OpenPiton's native coherence protocol to ARM's **CHI**, mapping its directory-based coherence engine onto CHI's **RN-F/HN-F/SN-F** node abstractions
- **RISC-V Training & Mentorship** — Conducted **student training programs** on RTL design and how to contribute to open-source
- **Programmable Permutation Engine** — Designed a 1024-bit programmable permutation [engine](#) [↗](#) at 300MHz clock and resolved Setup/hold violations for xc7z030ffg676-3 (Zynq-7000 series) FPGA

SOFT SKILLS

- **Languages** — English (C1), Urdu (Native)
- **Other** — Design review | Technical Documentation | Debugging & Root-Cause Analysis | Design verification coordination | Technical Leadership | Timing-Closure Problem Solving | Research & Mentorship | Volunteer work

PROJECTS

Google Summer of Code (GSoC), *Contributor (2022) / Mentor (2023)*

- Added **FuseSoC** support on the **Azadi-SoC** and **rebased** the **Ibex core** to latest upstream commits ([GSoC'22](#) [↗](#))
- Mentored **TileLink** Uncached Heavyweight (**TL-UH**) implementation in **Azadi-SoC** ([GSoC'23](#) [↗](#))

MARCore & NOVA1, *Research Lead / Contributor*

- Researched multicore cache-coherent memory systems using OpenPiton workflow and Ariane Core ([MARCore1](#) [↗](#))
- Contributed to NOVA1 SoC that contains RV64GC Core, UART, BRAM, DDR, and JTAG support ([The NOVA Project](#) [↗](#))

EDUCATION

2019 – 2023

Bachelor of Engineering in Electrical Engineering, *Usman Institute of Technology*

(CGPA: 3.289) (Specialization: Computer Systems)

- **Major Courses** — Digital Logic Design | Computer Architecture | Microprocessor-Based Systems | Embedded Systems | Operating Systems

Honors & Awards

- Hasham Foundation Scholarships by Usman Institute of Technology in 2022
- Linux Foundation (LiFT) Scholarship Recipient in 2024

CERTIFICATES

- **RISC-V Foundational Associate (RVFA)** — Date: Aug 2025 | Issuer: RISC-V International | Sponsor: Linux Foundation | [Link](#) [↗](#)
- **RISC-V Fundamental (LFD210)** — Date: Aug 2025 | Issuer & Sponsor: Linux Foundation | [Link](#) [↗](#)

PUBLICATIONS & ARTICLES

- Multicore SoC For SMP Linux ([Thesis](#) [↗](#))
- NOVA Emulation Project- An open-source RISC-V-based application-class AI SoC ([Project Doc](#) [↗](#))
- A Summer of Opportunity ([Medium — GSoC experience](#) [↗](#))
- Abstractions Can Lead Humanity to Ignorance (An Unpopular Opinion) ([Medium — technical opinion](#) [↗](#))